

Conditional Expectations of Exchangeable Summands

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Abstract

We construct an example of a symmetric random variable X such that the conditional expectation $\mathbf{E}[X \mid X + X' = 0]$ is undefined, even in cases where $\mathbf{E}[|X|] < \infty$.

Let $\{h_n\} \subseteq (0, \infty)$, $\{w_n\} \subseteq (0, \frac{1}{2})$ be non-negative so that $\sum_{n \geq 1} h_n \cdot w_n = \frac{1}{4}$, $\sum_{n \geq 1} h_n^2 \cdot w_n < \infty$. Defining

$$f(x) = \sum_{n \geq 1} h_n \cdot \{ \mathbf{1}_{|x-n| \leq w_n} + \mathbf{1}_{|x+n| \leq w_n} \},$$

let X denote a random variable with (symmetric) density f on $\mathbf{R}$. Since $w_n < \frac{1}{2}$, the supports of the indicator functions are disjoint for distinct n and opposite signs, which will aid our calculations.

Observe that one can sample from X by

1. $N \sim \mathbf{P}(N = n) = 4 \cdot h_n \cdot w_n$.
2. $S \sim \text{Uniform}(\{\pm 1\})$, $U \sim \text{Uniform}([-1, 1])$
3. Return $X = S \cdot N + U \cdot w_N$.

Consider taking two independent realisations of X and computing the conditional distribution of the first realisation, conditional on their sum. An intuitive symmetry argument would suggest that this distribution must have mean zero. Unfortunately, this mean does not exist.

Compute that $X + X'$ has a density at 0 given by $f_{[2]} = f \star f$, and hence that

$$\begin{aligned} f_{[2]}(0) &= \int_{\mathbf{R}} f(x)^2 \, dx \\ &= 4 \cdot \sum_{n \geq 1} h_n^2 \cdot w_n. \end{aligned}$$

As such, it follows that $[X \mid X + X' = 0]$ has a density given by

$$g(x) = \frac{f(x)^2}{f_{[2]}(0)}.$$

Check then that

$$\begin{aligned} \mathbf{E}[|X| \mid X + X' = 0] &= \int |x| \cdot \frac{f(x)^2}{f_{[2]}(0)} \, dx \\ &= \frac{1}{f_{[2]}(0)} \cdot \sum_{n \geq 1} h_n^2 \cdot \int_{|x-n| \leq w_n} 2x \, dx \\ &= \frac{4}{f_{[2]}(0)} \cdot \sum_{n \geq 1} n \cdot h_n^2 \cdot w_n \\ &= \frac{\sum_{n \geq 1} n \cdot h_n^2 \cdot w_n}{\sum_{n \geq 1} h_n^2 \cdot w_n}. \end{aligned}$$

Taking $h_n \sim n^\alpha$, $w_n \sim n^{-\beta} \cdot (\log n)^{-\gamma}$, check that if either

1. $\alpha \geq 0, \beta = 2 \cdot \alpha + 1, \gamma > 1,$
2. $\alpha \geq 0, \beta \in (2 \cdot \alpha + 1, 2 \cdot \alpha + 2), \gamma \geq 0,$ or
3. $\alpha \geq 0, \beta = 2 \cdot \alpha + 2, 0 \leq \gamma \leq 1,$

then it holds that $\sum_{n \geq 1} h_n \cdot w_n < \infty, \sum_{n \geq 1} h_n^2 \cdot w_n < \infty,$ but $\sum_{n \geq 1} n \cdot h_n^2 \cdot w_n = \infty,$ i.e.

$$\mathbf{E}[|X| \mid X + X' = 0] = \infty,$$

and hence that $\mathbf{E}[X \mid X + X' = 0]$ is undefined.

Check that

$$\mathbf{E}[|X|] = 4 \cdot \sum_{n \geq 1} h_n \cdot w_n \cdot n,$$

and this is seen to be finite if

1. $\alpha = 1, \beta = 2 \cdot \alpha + 1, \gamma > 1,$
2. $\alpha > 1, \beta = 2 \cdot \alpha + 1, \gamma > 1,$
3. $0 \leq \alpha \leq 1, \beta = \alpha + 2 \in (2 \cdot \alpha + 1, 2 \cdot \alpha + 2), \gamma > 1,$
4. $\alpha > 0, \beta \in (\max\{\alpha + 2, 2 \cdot \alpha + 1\}, 2 \cdot \alpha + 2), \gamma \in \mathbf{R},$ or
5. $\alpha > 0, \beta = 2 \cdot \alpha + 2, 0 \leq \gamma \leq 1.$

Moreover, if $\beta - 2 \cdot \alpha > 1$ or $\beta - \alpha > 2,$ then γ can be arbitrary, as the logarithmic terms are sub-dominant.

Visualisations

Figure 1: For $(\alpha, \beta, \gamma) = (0.5, 2.1, 0),$ plotting the density of $X.$

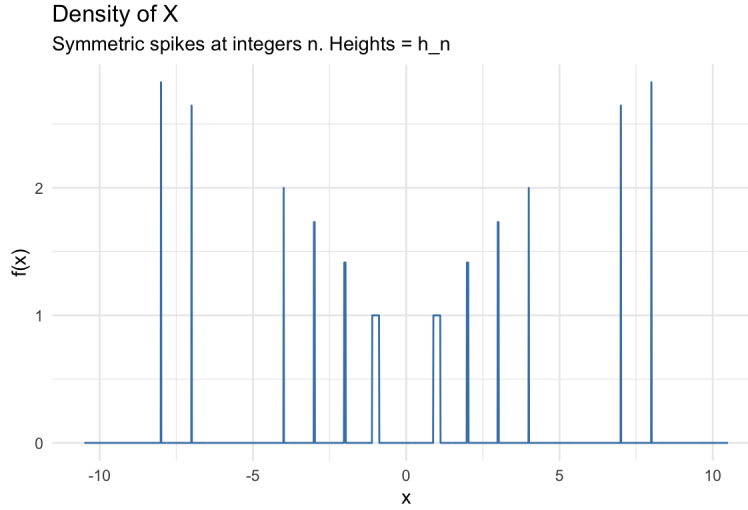


Figure 2: For $(\alpha, \beta, \gamma) = (0.5, 2.1, 0)$, plotting the approximate density of $X + X'$ by simulation.

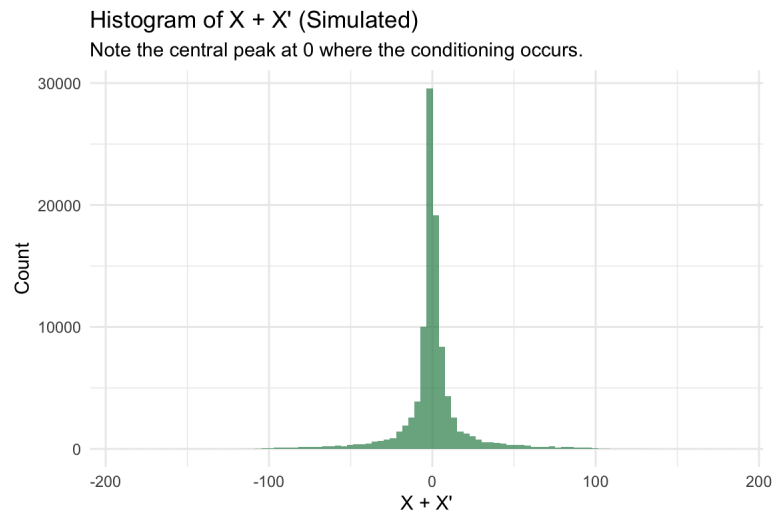


Figure 3: For $(\alpha, \beta, \gamma) = (0.5, 2.1, 0)$, plotting the conditional density of $X \mid X + X' = 0$.

